

REVIEW



# Tuning the ignition of CAR: optimizing the affinity of scFv to improve CAR-T therapy

Yanting Duan<sup>1,2,3</sup> · Ruoqi Chen<sup>1,2,3</sup> · Yanjie Huang<sup>4</sup> · Xianhui Meng<sup>1,2,3</sup> · Jiangqing Chen<sup>1,2,3</sup> · Chan Liao<sup>5</sup> · Yongmin Tang<sup>5</sup> · Chun Zhou<sup>6</sup> · Xiaofei Gao<sup>4</sup> · Jie Sun<sup>1,2,3</sup>

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## Abstract

How single-chain variable fragments (scFvs) affect the functions of chimeric antigen receptors (CARs) has not been well studied. Here, the components of CAR with an emphasis on scFv were described, and then several methods to measure scFv affinity were discussed. Next, scFv optimization studies for CD19, CD38, HER2, GD2 or EGFR were overviewed, showing that tuning the affinity of scFv could alleviate the on-target/off-tumor toxicity. The affinities of scFvs for different antigens were also summarized to designate a relatively optimal working range for CAR design. Last, a synthetic biology approach utilizing a low-affinity synthetic Notch (synNotch) receptor to achieve ultrasensitivity of antigen-density discrimination and murine models to assay the on-target/off-tumor toxicity of CARs were highlighted. Thus, this review provides preliminary guidelines of choosing the right scFvs for CARs.

**Keywords** Chimeric antigen receptor · scFv · Affinity · On-target/off-tumor · Adoptive cell therapy · Immunotherapy

## Introduction

Combining an extracellular antigen-recognition domain from antibodies with immune cell signaling domains, chimeric antigen receptor (CAR) can redirect T cell specificity and induce potent anti-tumor activity [1]. CAR-T therapy has achieved remarkable clinical success [1–4] and so far five products have been approved for clinical use by US Food and Drug Administration (FDA). Cytokine release syndrome (CRS) and neurotoxicity are common side effects caused by CAR-T therapy [5–7]. In addition, CD19 CAR-T cells were reported to cause B cell aplasia as normal B cells expressing CD19 also got eliminated by CAR-T cells [2, 8–10]. Although this on-target/off-tumor toxicity can be treated with immunoglobulin replacement therapy [2], this side effect can cause big problems in treating solid tumors since the tumor-associated antigens (TAAs) can be expressed in normal and vital organs [11, 12]. In renal cell carcinoma patients treated with carbonic anhydrase IX (CAIX) CAR-T, disorders of liver enzymes as well as autoimmune cholangitis were observed, resulting from the expression of CAIX on the bile ducts [13]. In an ERBB2-positive colon cancer patient, rapid respiratory failure developed after infusion of ERBB2 CAR-T cells and unfortunately the patient died from the injury of ERBB2-expressing lung epithelial cells [14].

Yanting Duan and Ruoqi Chen contributed equally to this work.

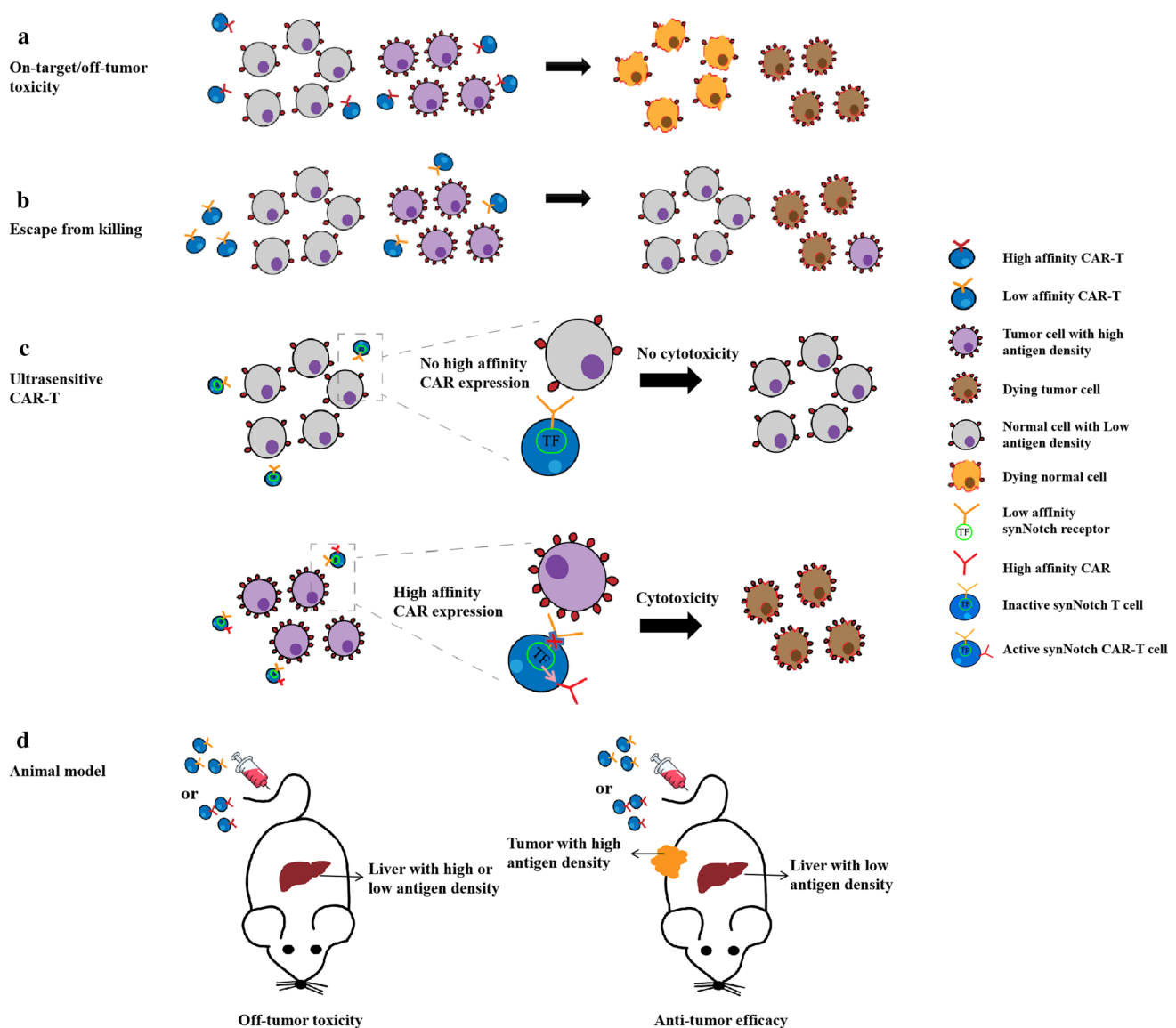
✉ Jie Sun  
sunj4@zju.edu.cn

- <sup>1</sup> Bone Marrow Transplantation Center of the First Affiliated Hospital & Department of Cell Biology, Zhejiang University School of Medicine, Hangzhou, Zhejiang, China
- <sup>2</sup> Liangzhu Laboratory, Zhejiang University Medical Center, Hangzhou, Zhejiang, China
- <sup>3</sup> Institute of Hematology, Zhejiang University & Zhejiang Engineering Laboratory for Stem Cell and Immunotherapy, Hangzhou, Zhejiang, China
- <sup>4</sup> Key Laboratory of Structural Biology of Zhejiang Province, School of Life Sciences, Westlake University, Hangzhou, Zhejiang, China
- <sup>5</sup> Department of Hematology-Oncology, Children's Hospital, Zhejiang University School of Medicine, Hangzhou, Zhejiang, China
- <sup>6</sup> School of Public Health, and Sir Run Run Shaw Hospital, Zhejiang University School of Medicine, Hangzhou, Zhejiang, China

Thus, different strategies to improve the safety of CAR-T therapy have been developed and summarized [11, 15].

As the antigen-recognition domain, single-chain variable fragment (scFv) initiates and determines the strength of T cell activation, providing specificity in an MHC-independent manner, which can prevent tumor escape through MHC downregulation [16]. Generally, CAR-T cells with a high-affinity scFv induces stronger anti-tumor activity than those with a low-affinity scFv [15, 17, 18], but also eliminate more normal cells with lower antigen density

(Fig. 1a). Indeed, several studies on CAR-T cells targeting CD19 [19], CD38 [20] HER2 [21, 22], GD2 [23], and EGFR [21, 24] have indicated that CAR-T cells constructed with low-affinity scFvs are less likely to trigger on-target/off-tumor toxicity without reduction of killing capability. In addition, synthetic biology approaches utilizing different affinities of scFvs to combat on-target/off-tumor toxicity have been developed [25]. Here, we summarize the efforts to improve CAR-T products through the optimization of scFvs.



**Fig. 1** Affinity-tuned CAR-T cells reduce the risk of on-target/off-tumor toxicity. **a** High-affinity CAR-T cells kill both tumor cells with high antigen density and normal cells with low antigen density; **b** tumor cells may escape from being killed by low-affinity CAR-T

cells; **c** ultrasensitive CAR-T cells based on the synthetic Notch (syn-Notch) system; **d** animal models to assess the risk of on-target/off-tumor toxicity of CAR-T

## scFv: a critical structural component of CAR

A CAR has a modular design with an extracellular antigen-recognition domain, a hinge, a transmembrane region and intracellular signal transduction domains [26]. The design of CAR has evolved several generations over the decades but an scFv derived from the antibody is used in all designs [27–30]. The scFv is a fusion protein of the variable regions of the heavy chain ( $V_H$ ) and light chain ( $V_L$ ) of monoclonal antibodies connected with a short linker peptide [31]. To choose a proper scFv for an antigen, antibodies with high binding affinity to the antigen are first selected and their  $V_H$  and  $V_L$  sequences are determined. Next, the scFv sequences are used to construct CARs specific for the antigen and in vitro killing assays are performed to screen the CARs with cytotoxic functions. Not many scFvs pass this test as the stability and solubility of a scFv as well as how exposed the scFv's binding epitope on the target cells all affect the functionality of the CAR. Only limited numbers of CARs with high cytotoxicity are selected for in vivo validation using mouse models and even fewer CARs with potent anti-tumor activity are chosen for potential clinical use.

With the discovery that TCR affinity correlates tightly with the efficiency of cellular activation [32], the affinity of scFv fragments has also been found to influence the activation of T cells [33]. CAR-T cells with high-affinity scFvs can trigger strong T cell activation and effective lysis of target cells, but an activity threshold for scFv affinity may

exist, and beyond it, there is no additional increase in CAR-mediated cell activation [33, 34]. In addition, scFv affinity can influence the density discrimination of CAR-T cells, as several studies have suggested that the optimal affinity of scFv enables selective killing of CAR-T against tumors with high antigen density, thus reducing the on-target/off-tumor adverse effects [20, 21, 24].

To determine which scFvs are suitable to construct CAR, several technologies are commonly used to measure the affinity of antibodies/scFvs, such as Enzyme Linked Immunosorbent Assay (ELISA), Fluorescence-Activated Cell Sorting (FACS), Surface Plasmon Resonance (SPR) and Biolayer Interferometry (BLI) [35]. Both ELISA and FACS measure the parameter  $EC_{50}$  (concentration for 50% maximal effect) while SPR and BLI measure the parameter  $K_D$  by monitoring the real-time binding kinetics of antigen and antibody [35]. The lower the  $EC_{50}$  or  $K_D$ , the higher the binding affinity. Even though BLI assays are high throughout, SPR measurement is more reproducible with a continuous flow system. Notably, SPR and BLI as well as ELISA assay all use soluble antigens for antibody binding measurement while in FACS assay antigens are expressed on cell surface. Therefore, it is possible that FACS assays are more suitable to screen scFvs for CAR construction than other assays as CARs also bind to antigens expressed on cell surface. In this review, respective assays measuring affinities of scFvs for different antigens summarized in Table 1 have been indicated.

**Table 1** scFv in CAR with different affinity (SPR, BLI or FACS results)

| Target                 | 0.1–1 (nM)    | 1–10 (nM)                    | 10–100 (nM)                                           | 100–1000 (nM)              | 1000–5000 (nM)                            | Other                     |
|------------------------|---------------|------------------------------|-------------------------------------------------------|----------------------------|-------------------------------------------|---------------------------|
| CD19 ( $K_D$ [19])     | FMC63 (0.328) |                              | CAT19 (14.38) <sup>a</sup>                            |                            |                                           |                           |
| CD38 ( $K_D$ [20])     |               | 028 (1.8)                    | A1 (17)                                               |                            | A4 (1915) <sup>a</sup>                    | B1 (N.A.)<br>B3 (N.A.)    |
| HER2 ( $K_D$ [21, 22]) | 4D5 (0.58)    | 4D5-7 (3.2)                  |                                                       |                            | 4D5-5 (1119) <sup>a</sup><br>4D5-3 (3910) |                           |
| GD2 ( $K_D$ [23])      |               | 4D5_WT_highest (1.9)         | 4D5_high (17.6)<br>4D5_medium (46.5)                  | 4D5_low (210) <sup>a</sup> |                                           |                           |
| EGFR ( $EC_{50}$ [21]) | 2224 (0.94)   | 3F8 (5)                      | 14G2a (77)<br>p2-4 (15.4)<br>p3-5 (88.2) <sup>a</sup> | c10 (263.7) <sup>a</sup>   |                                           |                           |
| EGFR ( $K_D$ [25])     |               | Anti-EGFR-highest (2.8)      | Anti-EGFR-high (10.4)                                 |                            | Anti-EGFR-low (3800) <sup>a</sup>         | Anti-EGFR-lowest (50,000) |
| EGFR ( $K_D$ [24, 67]) |               | Cetuximab (1.8) <sup>b</sup> | Nimotuzumab (21) <sup>a,b</sup>                       |                            |                                           |                           |

N.A. not applicable

<sup>a</sup>Optimized choice

<sup>b</sup>Affinity of the full-length antibody

## Tuning the affinity of scFv to optimize CAR-T cell function

TAAAs in solid tumors can be distributed in vital organs and normal tissues, but antigens in hematological malignancies can usually be restricted to cells of hematological lineage [36], which limits the scope of the on-target/off-tumor toxicity. Thus, we discuss the impact of tuning the affinity of scFv of CAR-T for hematological malignancies and solid tumors separately.

## Tuning the affinity of scFv to treat hematological malignancies

### CD19

CD19 is the earliest and most widely used target in CAR-T immunotherapy, which has been chosen because of its expression on both malignant and normal B-lineage cells, but rarely on other normal tissues or hematopoietic stem and progenitor cells [36]. Currently, all four commercial CD19 CAR-T products use the same scFv derived from a murine monoclonal antibody (mAb), FMC63, which has a relatively high affinity against CD19 ( $K_D = 0.328$  nM, all affinities listed below are  $K_D$ , unless otherwise noted) [37]. A study of a novel CD19 CAR (CAT19, 14.38 nM) with affinity > 40-fold lower than FMC63 (Table 1) [19] showed that surprisingly CAT CAR-T cells exhibited significantly greater cytotoxicity and proliferation than FMC63 CAR-T cells in vitro [19]. In a xenograft NALM-6 model, CAT CAR-T cells exhibited a significantly higher anti-tumor activity and enhanced proliferation with higher expression of CD127 (IL7-R $\alpha$ ) and anti-apoptotic B cell lymphoma-2 (Bcl-2) than FMC63 CAR-T cells. Meanwhile, there were no significant differences in the expression of activation and exhaustion markers (TIM-3, LAG-3, PD-1) [19]. Based on this, a clinical study (CARPALL, NCT02443831) used CAT CAR-T cells to treat patients with advanced relapsed/refractory pediatric B cell acute lymphoblastic leukemia (ALL) [19]. One-year overall (OS) and event-free survival (EFS) was 63% and 46%, comparable with the 76% and 50% in the ELIANA trial of tisagenlecleucel (an FMC63-based CAR-T cell product). Importantly, the CRS with CAT CAR-T was mild and no severe grade 3/4 CRS was observed in any of the patients. In comparison, 46% patients of the ELIANA study experienced grade 3/4 CRS and 47% patients were admitted to intensive care unit.

Utilizing transcriptomic analyses (RNA-seq) and mass cytometry analyses (CyTOF), the team further explored the molecular mechanisms that contribute to the functional

differences between CAT and FMC63 CAR-T [38]. The results indicated that CAT CAR-T cells were more likely to exhibit a polyfunctional pattern of cytokine expression, with a marked increase in the proportion of cells co-expressing three or more effector or stimulatory cytokines [38]. Moreover, CAT CAR-T cell displayed increased levels of trogocytosis [38], a phenomenon involving the transfer of antigen from tumor cells to CAR-T cells (trog<sup>+</sup> CAR-T) [39]. Interestingly, trog<sup>+</sup> CAT CAR-T exhibited longer persistence and retained robust activation even after the clearance of tumor cells, which may help to prevent relapse [38]. The authors suggested that CAT CAR-T with reduced scFv affinity preferentially recognized tumor cells with high CD19 expression rather than mediating fratricide while the trogocytosed CD19 protein may provide sustained signaling [38]. Thus, the affinity-tuned CAR-T may have the potential to enhance both safety and efficacy in clinical setting.

### CD38

CD38, a type II transmembrane glycoprotein, is an attractive target due to its high expression in various hematological malignancies especially multiple myeloma (MM) [40]. Recently, several types of solid tumors including the oral squamous cell carcinoma, nasopharyngeal carcinoma and cervical carcinoma were shown to express CD38 [41–43]. The FDA-approved CD38 targeting therapeutic mAb daratumumab (DARA) has been used for the treatment for MM [44] and CD38 CAR-T therapy has also been proven effective in both preclinical studies and clinical trials [40]. However, decreased percentages of mononuclear cells and granulocytes were reported in one MM patient treated with CD38 CAR-T cell therapy [45] as CD38 is also expressed at low levels on normal precursor B cells, NK cells, and myeloid precursors [46].

Drent and colleagues generated scFvs from antibodies with approximately 1000-fold difference of affinities to CD38 (Table 1) via light-chain exchange [20]. Four CARs with lower affinities (A1, 17 nM; A4, 1915 nM; B1, N.A.; B3, N.A.) (Table 1) than 028 control (1.8 nM) showed anti-MM activity in vitro, among which B1 and B3 CAR-T secreted lower level of cytokines than the control [20]. Meanwhile A4, B1 and B3 CAR-T exhibited a similar or higher growth rate than that of 028 and A1 CAR-T cells. Thus, the A4 CAR-T was selected as the best low-affinity CD38 CAR-T to be compared with 028 in vivo. Interestingly, A4 and 028 CAR-T cells exhibited similar anti-tumor effect in a xenograft model, but A4 caused lower toxicity on CD38<sup>+</sup>CD34<sup>+</sup> hematopoietic progenitor cells and negligible toxicity on CD38<sup>+</sup> fractions of normal blood cells, such as B, T and NK cells. Of note, another problem of CD38 CAR-T cells is that they can kill each other (fratricide) due

to the expression of CD38 in T cells [20, 47], which had also been reported by Drent's study [20]. Compared to the high-affinity 028 CAR-T cells which lost CD38 expression after 2 weeks' in vitro culture, most low-affinity CD38 CAR-T cells retained CD38 expression and had a similar or better growth rate, indicating a reduced fratricide [20]. Furthermore, their subsequent study showed that the preferential recognition of myeloma cells by affinity-optimized CD38 CARs is also dependent on their co-stimulatory domain. If the CD28 co-stimulatory domain is used, the affinity can be even much lower ( $K_D < 1900$  nM) [48].

The promising results of tuning the scFv for both CD19 and CD38 CARs taught us that it is worthwhile to do the same to CARs targeting other well-established hematological antigens, such as BCMA, CD20, CD22, in preclinical models and clinical studies. For CAR-T targeting CD5, CD7, or other T cell expressing antigens, it is possible that scFv affinity optimization can be used in combination with CRISPR/Cas9-mediated gene editing or other blockers to further reduce fratricide [49].

### Tuning the affinity of scFv for treating solid tumors

Various strategies have been used to increase the specificity of CAR-T cells against solid tumors. For example, CAR-T cells get activated when both tumor antigen A and B are present [50]; CAR-T cells be activated only when tumor antigen A but not normal antigen B is present [51]. However, these multi-antigen-targeted CAR-T cells are only proven effective in preclinical models but have limited applications in clinical trials [11]. Meanwhile, recent studies have found that simply modifying the affinity of scFv in CAR-T can effectively reduce the on-target/off-tumor effect of CAR-T while maintaining its killing capability [21, 22, 24]. This provides an alternative approach to optimize CAR-T to overcome on-target/off-tumor toxicity.

#### HER2

As one of the most studied TAAs, human epidermal growth factor receptor 2 (HER2) is overexpressed in many tumors such as breast cancer, ovarian cancer and glioblastoma multiforme (GBM) [18, 21]. Herceptin (trastuzumab), a mAb specifically binding to HER2, has been approved for the treatment of early-stage HER2-positive breast cancer [21]. Recently, several studies showed that solid tumors such as osteosarcoma and breast cancer resistant to trastuzumab can still be eliminated by CAR-T cells redirected by the same antibody domain [52, 53]. Notably both trastuzumab and HER2 CAR-T cells can induce serious side effects such as the cardiotoxicity and lung damage, linked to the presence of the HER2 on the surface of cardiomyocytes and lung epithelium [14, 54].

In 2015, Liu et al. [21] found that targeted mutation of Herceptin (4D5, 0.58 nM) resulted in scFvs with reduced affinities, namely 4D5-7, 4D5-5 and 4D5-3 with  $K_D$  being 3.2 nM, 1119 nM, and 3910 nM, respectively (Table 1). 4D5-7 CAR-T exhibited strong cytotoxicity to tumor cells expressing both high and low levels of HER2, while 4D5-5 or 4D5-3 CAR-T cells were sensitive to high but not low HER2-expressing tumor lines in vitro. Additionally, 4D5-5 CAR-T can effectively inhibit the growth of high but not low HER2-expressing tumors in vivo. Thus, reducing the affinity of scFv for HER2 from 0.58 to 1119 nM (approximately 1930-fold) is a practical solution to eliminate the on-target/off-tumor toxicities [21]. Another study constructed high and low HER2-expressing liver models in mice to evaluate the on-target/off-tumor toxicity [22] (Fig. 1d). They found that 4D5 HER2 CAR-T exhibited high activity to both high and low HER2-expressing livers while 4D5-5 CAR-T cells reduced liver damage on HER2-low livers. When HER2-high SKOV3 tumor cells were implanted subcutaneously in mice with HER2-low livers which can better mimic the normal tissues or organs with low antigen density in human, 4D5-5 CAR-T cells had better tumor control. Detailed study revealed that HER2 CAR-T with different affinities first aggregated in HER2-low livers, but the 4D5-5 CAR-T exited the liver and infiltrated into tumor more rapidly than the 4D5 CAR-T, causing earlier tumor regression [22].

#### GD2

Disialoganglioside (GD2) is a surface glycolipid antigen overexpressed in tumors including neuroblastoma, melanoma, retinoblastomas and Ewing's sarcoma but have restricted expression in normal tissues [55, 56]. Although several anti-GD2 mAbs have been developed to target solid tumors especially neuroblastoma, clinical trials found limited responses and nearly half of the patients with neuroblastoma would relapse [57, 58]. Moreover, the infusion of FDA-approved anti-GD2 mAb dinutuximab causes transient neuropathic pain-like syndrome [59]. GD2 CAR-T cells have been used to achieve a better anti-tumor effect than mAbs [57].

Richman et al. [23] constructed GD2 CAR-T cells using 14G2a-derived scFv (77 nM), confirming their anti-tumor activity in vitro but not in vivo, probably for the reason that the affinity of scFv was not high enough to trigger strong anti-tumor response (Fig. 1b, Table 1). Proposing that the short linker (only nine amino acids) between the variable domains of the scFv might affect its stability, the author lengthened the linker to 20 amino acids but found no improvement in the cytotoxicity of CAR-T cells in vitro. Next, they introduced an E101K point mutation into the 14G2a scFv to generate a higher affinity GD2 CAR. In a neuroblastoma xenograft model, these GD2<sup>E101K</sup> CAR-T



cells displayed enhanced anti-tumor activity, concomitant with severe central nervous system (CNS) toxicity and rapid mortality, which has also been reported in another study using 3F8 scFv (5 nM) (Fig. 1a, Table 1) [23]. CAR-T cell infiltration in the cerebellum and basal regions of the brain with low expression of GD2 can also be observed [23]. Thus, other components such as the hinge domain and the co-stimulatory domain need to be optimized as well. Two recent studies [60, 61] reported the choice of co-stimulatory domain of CAR played an important role in the on-target/off-tumor toxicity. GD2<sup>E101K</sup> CAR-T cells with CD28 co-stimulatory domain but not 4-1BB domain reported no neurotoxicity symptoms in a xenogeneic model [61]. Other factors including the choice of gene-transfer vector and even the culture conditions of T cells may also influence the toxicity [60].

## EGFR

Epidermal Growth Factor Receptor (EGFR/HER1), associated with tumor reoccurrence, neoangiogenesis and metastases, is overexpressed in breast cancer and glioblastoma [62]. EGFR CAR-T cells have been demonstrated effective in several clinical trials [63–65], but with on-target/off-tumor toxicity on normal cells, especially epithelial cells [66]. Two CARs were generated from cetuximab (1.8 nM, affinity of the full-length antibody) and nimotuzumab (21 nM, affinity of the full-length antibody), two FDA-approved therapeutic humanized mAbs against EGFR with tenfold difference in affinity (Table 1) [24, 67]. Nimotuzumab CAR-T cells produced less IFN- $\gamma$  than cetuximab CAR-T while they had similar CAR expression and proliferation [24]. In vitro study showed that cetuximab CAR-T cells killed glioma cells regardless of EGFR density while nimotuzumab CAR-T selectively lysed cells of high EGFR expression (Fig. 1a) [24]. In vivo both cetuximab and nimotuzumab CAR-T cells can inhibit the growth of high EGFR-expressing glioma, while only cetuximab CAR-T cells can control the growth of low EGFR-expressing glioma [24]. Meanwhile, rapid weight loss and lethargy were only observed in mice treated with cetuximab CAR-T cells. Thus, nimotuzumab CAR-T cells might have the potential to discriminate cells with different EGFR densities, which was described as “discriminate friend from foe” [24].

Similar results were also observed in another study [21] developing a panel of CARs with diverse affinities of anti-EGFR scFvs, namely 2224 ( $EC_{50}$  = 0.94 nM), P2-4 ( $EC_{50}$  = 15.39 nM), P3-5 ( $EC_{50}$  = 88.24 nM), and C10 ( $EC_{50}$  = 263.67 nM) (Table 1). High-affinity 2224.BBZ and P2-4.BBZ CAR-T cells exhibited strong anti-tumor activity to all EGFR<sup>+</sup> tumors regardless of EGFR expression level. In contrast, low-affinity P3-5.BBZ and C10.BBZ CAR-T

cells exhibited a much weaker response to tumor cells of low EGFR expression than those of high EGFR [21].

Taken together, studies of CARs targeting CD19, CD38, HER2 or EGFR showed that an appropriate reduction in scFv affinity might not significantly reduce the killing activity of CAR-T, but at the same time could effectively attenuate on-target/off-tumor effects. We summarize that other than GD2, the optimal affinity of scFvs targeting antigens described above (annotated with <sup>a</sup>) is generally in the range of 10–5000 nM (Table 1). For GD2 CAR, tuning the affinity alone is not enough thus more strategies are needed to achieve both safety and efficacy [60, 61].

## Synthetic biology approach in combination with scFvs of different affinities

Currently, synthetic biology-based strategies are explored to improve the efficacy and safety of therapeutic T cells [68–70]. The synthetic Notch (synNotch) system [71] contains a receptor that can recognize one tumor antigen through scFv and then activate a fused transcription factor (TF) to induce the expression of a transgene, such as a CAR gene [71]. Based on this “AND” logic gate, CAR-T cells can only be triggered in the presence of two tumor antigens, increasing the specificity of CAR-T cells [70]. Recently, Rogelio and colleagues developed a new use of synNotch system to discriminate cells with different antigen densities [25]. They first mutated the anti-HER2 Herceptin (4D5, 1.9 nM) to make three scFv variants, with, respectively, high (17.6 nM), medium (46.5 nM) or low (210 nM) affinity. Next, they constructed three synNotch receptors and tested their sensitivities to antigen density. Only the synNotch receptor with a low-affinity scFv (210 nM) can act as the circuit filter to achieve ultrasensitivity [25] and thus was chosen to induce the expression of a CAR with high-affinity scFv (17.6 nM) (Fig. 1c). In vitro killing assays showed this design could discriminate cell lines of diverse HER2 densities, killing high HER2 density tumors while sparing low HER2 density cells. Interestingly, delayed onset of tumor killing was observed probably because additional time is needed to induce sufficient high-affinity HER2 CARs after synNotch activation. Strikingly, synNotch HER2 CAR-T cells cleared the high-density tumor efficiently while they fail to inhibit the low-density tumor in the same mouse [25]. This two-step circuit approach was also validated against EGFR where synNotch with a low-affinity scFv (3800 nM) acted as the antigen-density filter to activate the expression of a high-affinity (10.4 nM) EGFR CAR [25]. Of note, the affinity difference in EGFR synNotch and CAR was approximately 365-fold while that in the HER2 setting was only 12-fold, indicating the working high and low affinities against different targets need to be optimized.

Several questions remain for this innovative design. First, after the induction of high-affinity CAR, can these cells cause damage to normal cells? Second, are the scFvs used in the synNotch HER2 CAR study optimal? As another study [21] noted that HER2 CAR-T with even lower affinity (1119 nM) scFv were still able to achieve strong killing of high HER2-expressing tumor while reducing the off-tumor toxicity on low HER2-expressing tissues, can we increase the affinity differences of anti-HER2 scFv to yield a more significant result? Nevertheless, this approach provides alternative solutions for on-target/off-tumor toxicity.

### Animal models to assess the safety of CAR-T cells

Mouse models have been established to reveal the mechanisms of CRS related to CAR-T therapy [72, 73], but there is a lack of animal models to assess the on-target/off-tumor effects. Owing to the differences in antigen expression pattern and level between mice and human as well as cross-species reactivity of the scFv/antibody, most preclinical studies have not validated the off-tumor toxicity of CAR-T cells, which would increase the risk of participants in clinical trials [13, 14].

To investigate the potential off-tumor effect on CD38<sup>+</sup> normal hematopoietic progenitor cells, Drent and colleagues used RAG2<sup>-/-</sup>γc<sup>-/-</sup> immunodeficient mice with reconstructed human hematopoietic niche [20]. In detail, subcutaneous implantation of ceramic scaffolds coated with human bone marrow stromal cells created humanized bone marrow-like niches where fluorescently labeled (FarRed dye) CD38<sup>+</sup>CD34<sup>+</sup> normal hematopoietic progenitor cells were injected. The fluorescence lifetime imaging (FLI) signal from these normal hematopoietic progenitor cells were measured in vivo at week 1 and 3 after intravenous injection of high-affinity 028 or low-affinity A4 CD38 CAR-T cells. They showed that live FLI signals of these mice groups had no difference. As the fluorescence of FarRed-labeled cells decreases with each division, this assay may have limited sensitivity. Therefore, the total cell numbers and percentage of CD38<sup>+</sup> cells within the CD34<sup>+</sup> fraction were also measured by flow cytometry in the post-mortem analysis [20]. Mice injected with the low-affinity A4 CAR-T showed both higher trend of CD38<sup>+</sup> percentage and cell numbers within the CD34<sup>+</sup> fraction compared to mice treated with 028 CAR-T cells, indicating low-affinity CAR-T cells might have less off-tumor activity on CD34<sup>+</sup> progenitor cells [20].

Recently, another preclinical model was developed by Castellarin et al. to assess the on-target/off-tumor toxicity in the solid tumor context (Fig. 1d) [22]. Murine livers expressing different levels of human HER2 were constructed using AAV gene delivery or transgene transposition by the piggy-Bac transposon system. The injury of CAR-T cells to liver due to the on-target/off-tumor toxicity can be measured by

indexes such as the elevation of serum alanine aminotransferase levels and the liver pathology, and in vivo BLI can be used to track the migration of CAR-T in vivo. Researchers pointed out that the liver might be suitable for testing the toxicity of CAR-T for its high perfusion and easy exposure to antigens, which can better mimic the physiological level of antigen expression in humans [22]. This model enabled the interesting discovery that both high- and low-affinity CAR-T cells first aggregated in the liver with low HER2 expression. However, low-affinity CAR-T cells moved from liver to high HER2-expressing tumor faster than high-affinity cells, which may contribute to its better anti-tumor efficacy. This study focuses only on liver toxicity but opens door to establish future animal models with other organs/tissues expressing targeted antigens. Above all, preclinical models to evaluate the density discrimination of CAR-T cells are still limited and need further development.

### Conclusions

After summarizing several studies, we found that tuning affinity of scFv is a practical way to reduce the on-target/off-tumor toxicity of CAR-T therapy. CAR-T with high-affinity scFvs generally failed to discriminate between high-density tumors and low-density normal tissues while scFvs with low affinity might be suitable for constructing optimal CARs (Fig. 1). Interestingly, appropriately reducing the scFv affinity may retain the anti-tumor capacity of CAR-T, probably because of the “affinity threshold” or “affinity ceiling” of CARs [19, 21, 24, 33]. An increase in the affinity of scFv above this threshold will not result in an enhancement of CAR-T activation and the killing [33]. This threshold is influenced by a number of factors including the antigen density on the surface of target cells, CAR expression levels as well as the binding epitope localization [74]. However, affinity-optimized CARs may have potential tumor escape issue as target antigen expression levels can decrease to normal levels or even lower on tumor cells through clonal selection or therapy-induced downregulation of the antigen (Fig. 1) [75–78]. Of note, most studies described above detected the affinity of scFvs or antibodies in a soluble form at 3D level, which is suitable for antibodies as they are naturally soluble. However, this measurement method of a scFv may not be ideal to predict the functionality of the CAR constructed from the scFv, as the scFv and CAR will be displayed on the cell membrane and actually interacts with its target antigen at 2D level. Therefore, we suspect that the affinity detection of scFv at 2D level would be more appropriate for CAR construction [79].

Nevertheless, tuning the scFv affinity of CARs alone may not be enough to solve the on-target/off-tumor side effect. Other features such as the structure of scFv [80], the length

of hinge [17], the choice of co-stimulatory domain [48, 81] and the strength of activation domain [82] also need to be taken into account to construct an optimal CAR with balanced safety and efficacy. Last but not least, other strategies such as synthetic biology approaches can be combined with scFv affinity optimization to enhance the performance of CAR-T cells in cancer immunotherapy.

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## Declarations

**Conflict of interest** The authors declare that they have no competing interests.

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